

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459908.727 +/- 0.025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.023
Transit depth (Rp/Rs)^2	1.48 +/- 0.57 [%]
Orbital Inclination (inc)	89.67 +/- 2.03
Airmass coefficient 1 (a1)	1.23 +/- 0.012
Airmass coefficient 2 (a2)	-0.131 +/- 0.013
Scatter in the residuals of the lightcurve fit is	0.93 %
Best Comparison Star	None
Optimal Aperture	6.06
Optimal Annulus	8.59
Transit Duration (day)	0.081 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.122 $\pm$ 0.023 vs 0.1404 (deviation 0.8 σ)
Duration fit vs published	0.081 d vs 0.0754 d (deviation 0.37 σ)
Mid-time O–C	47.0 min
Depth SNR	5.3
Residual scatter	0.93 %

Light curve

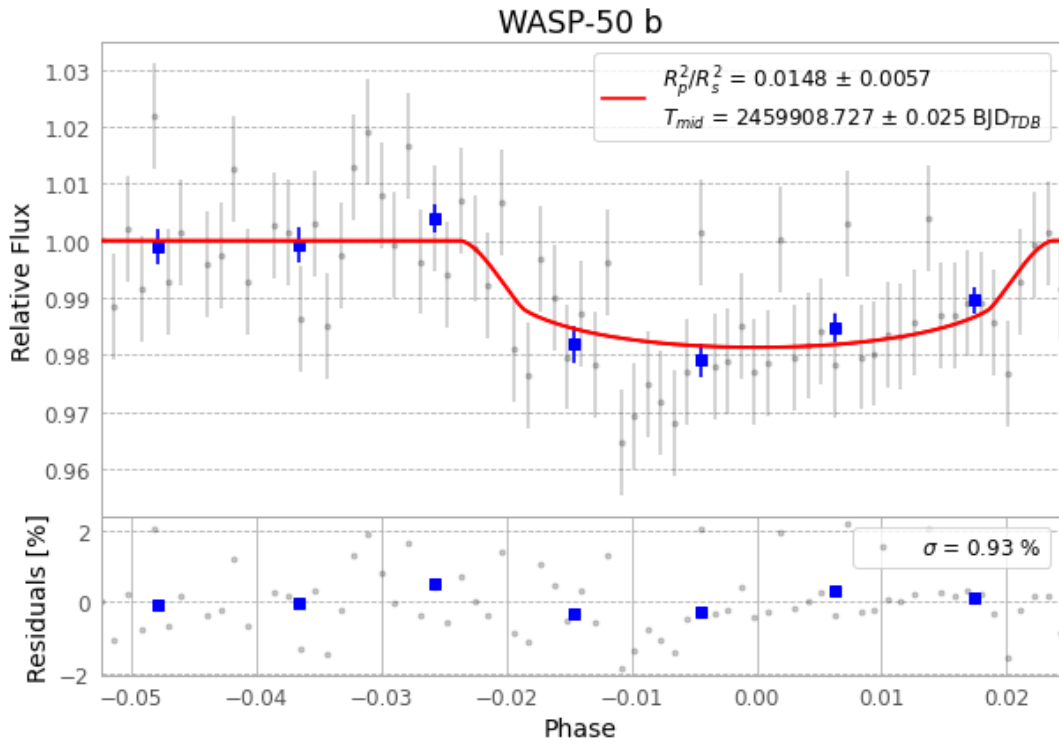


Figure 1 — FinalLightCurve_WASP-50 b_2022-11-24.png (SHA-256 6776ce974d422daf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [308, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 20af4995b5e46ddb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

73 FITS frames (48 MB), WASP-50221125025115.FITS → WASP-50221125062715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-221125.log (4f6760cb6f67...), FinalLightCurve_WASP-50 b_2022-11-24.png (6776ce974d42...), FinalLightCurve_WASP-50 b_2022-11-24.csv (0b0462a4df07...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 05:14Z.